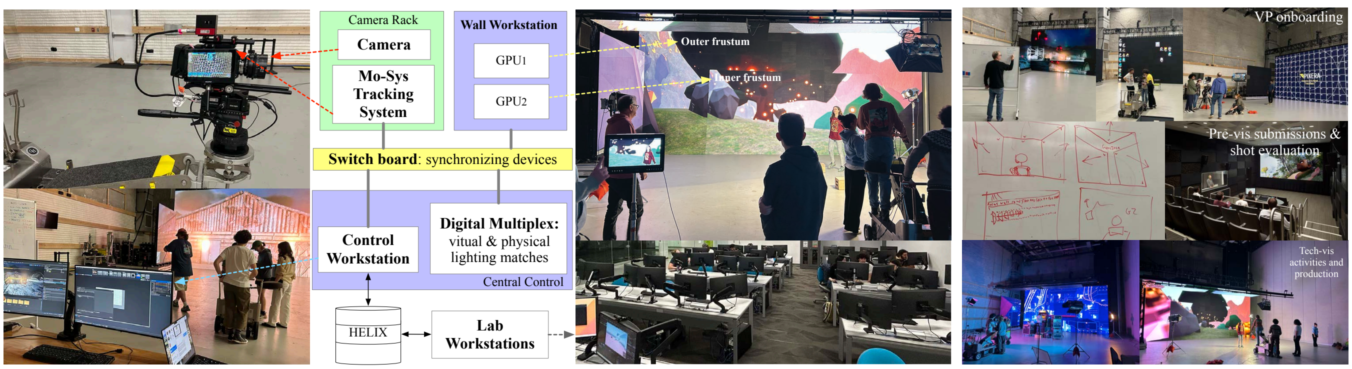
Rolling Through the Pipeline: A Course Model for Virtual Production Education

Flip Phillips
fxpppr@rit.edu
Rochester Institute of Technology
Rochester, New York, USA

David Long
dllppr@rit.edu
Rochester Institute of Technology
Rochester, New York, USA

Aidan Montag
ajm3312@rit.edu
Rochester Institute of Technology
Rochester, New York, USA

Chao Peng
cxpigm@rit.edu
Rochester Institute of Technology
Rochester, New York, USA



(a) Overview of the virtual production platform **(b)** Examples of class activities

Figure 1: An overview of the virtual production platform used for the course and the examples of class activities.

ABSTRACT

This paper presents an introduction-level virtual production (VP) studio course. The course integrates creative and technical domains across film production and dynamic scene generation to simulate an industry-grade collaborative workflow. Through project-based learning and cross-disciplinary collaboration, students develop both production literacy and creative confidence. This paper outlines the instructional framework, learning outcomes, and challenges of fostering technical and narrative coherence in VP education.

CCS CONCEPTS

• **Social and professional topics** → **Computing education programs**.

ACM Reference Format:
Flip Phillips, David Long, Aidan Montag, and Chao Peng. 2026. Rolling Through the Pipeline: A Course Model for Virtual Production Education. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Educator's Forum (SIGGRAPH Educator's Forum '26), July 19–23, 2026, Los Angeles, CA, USA.* ACM, New York, NY, USA, 3 pages. https://doi.org/10.1145/3799829.3812522

SIGGRAPH Educator's Forum '26, July 19–23, 2026, Los Angeles, CA, USA

ACM ISBN 979-8-4007-2552-4/2026/07
https://doi.org/10.1145/3799829.3812522

1 INTRODUCTION

By merging real-time computer graphics with cinematic practice, virtual production (VP) transforms live-action filmmaking from a linear pipeline into an interactive, collaborative workflow where creative decisions, technical problem-solving, and aesthetic refinement occur simultaneously [Bennett et al. 2023]. Learning VP helps students understand how artistic intent is translated into computational systems through game engines, LED stages, and camera tracking [Perkins and Echeverry 2022; Swords and Willment 2024]. The integration fosters a new form of visual literacy that combines creative fluency with technical adaptability, preparing students across film, animation, game design, and media technology for evolving professional practices.

Existing VP curricula often emphasize tool-specific training [Kavakli and Cremona 2022], or treat VP as an extension of animation or game design courses [Maraffi 2024; Najafi et al. 2024]. As more institutions invest in VP facilities, there is a growing need for pedagogical approaches that reflect integrated workflows aligned with real-world production [Boutellier and Raptis 2023].

This paper presents the design, implementation, and pedagogical reflections of an introductory VP course. The course emphasizes how creative intent propagates through version-controlled digital workflows, real-time engines, stage hardware, and on-site cinematography under live production constraints. The course demonstrates how industry-standard tools, systems, and pipelines can be adapted for early VP education while balancing creative exploration with technical rigor and preserving the flexibility (e.g., pacing and learning styles) of an academic environment.

2 INDUSTRY-ALIGNED PLATFORM

Our VP course is supported by an LED-wall-based stage that displays virtual environments, providing perspective-correct imagery for the camera while immersing performers within the scene. The immediacy of this visualization ($32' \times 16'$ LED wall with 2.3 mm pixel pitch) turns the stage into an active learning space where students directly observe how artistic intent affects system behavior.

The LED display system is driven by a processor rack with dual NVIDIA A6000 GPUs that render separate inner and outer views, which are sent to Brompton processors controlling the LED panels. The inner view matches the camera's frustum, while the outer view provides surrounding context for lighting, reflections, and environmental continuity, mirroring common industry VP practice.

A central workstation runs Unreal Engine, manages color calibration, and coordinates communication across the camera rack, lighting controls, and lab workstations over an Ethernet network. The camera rack uses a Mo-Sys tracking system to synchronize the physical and virtual cameras, ensuring accurate alignment between the LED display and camera projection.

3 COURSE MODEL

The primary objective is to develop both a conceptual and operational understanding of the VP platform in practice. The course helps students understand how early creative decisions propagate through a VP workflow and shape downstream technical outcomes.

The course assumes a heterogeneous student population from film, animation, game design, and digital media backgrounds. While formal prerequisites are minimal, familiarity with imaging concepts and at least one production tool (e.g., game engine, 3D modeling, or version control) is recommended. A key goal is to build confidence in multi-role collaboration.

The course uses three specialized learning environments. Conceptual critique occurs in a **screening room**, where students evaluate shot planning, camera logic, and narrative flow. Digital asset creation takes place in a **media laboratory** equipped for modeling, animation, editing, and version control. Full hardware integration happens on the **LED-wall stage**, where students work with displays, tracking systems, lighting, and networked devices, confronting practical VP challenges such as calibration, spatial layout, and system alignment during live testing.

3.1 Learning Structure and Class Activities

Instruction begins with a system-level introduction to the VP ecosystem. Students tour the LED-wall stage to examine rendering pipelines, camera–tracking, and networked production systems, paired with analysis of professional VP examples and hands-on exploratory sessions operating the camera and wall. These embodied activities, spanning the first two weeks, ground abstract concepts in physical and real-time production practice.

At the same time, students develop individual concept ideas through storyboards, engine blockouts, and visual sketches. These ideas are reviewed for VP feasibility, refined through critique, and gradually converge into one or two shared directions. Around weeks 4-5 of the semester, students form production teams based on their emerging interests in these directions.

Once teams are established, students move between the media lab and stage, iteratively testing camera placement, lighting, spatial composition, and virtual–physical alignment. Through these cycles of design and staged experimentation, they learn how early creative decisions shape downstream technical constraints.

As projects progress, students rotate across production roles, including performer, camera operator, lighting technician, and control operator, gaining practical insight into the coordination required for VP workflows.

3.2 Assignment and Assessment Framework

Assessment follows three milestones aligned with the VP pipeline. The **storyboard assignment** establishes the conceptual foundation, where students define narrative flow, shot composition, and staging intent. **Previsualization** uses a rolling submission model in which students iteratively share shot drafts, camera tests, and environment sketches, allowing early feedback and identification of potential production issues. The **technical visualization** milestone formalizes the production plan, detailing toolchains, assets, role assignments, and stage readiness. Evaluation across all milestones balances creative clarity, technical planning, and collaboration, with attention to both team outcomes and individual contributions.

4 RESULTS AND TAKEAWAYS

The course enrolled 11 students from film, animation, visual communication, and game design, and met twice weekly for three-hour sessions. Students were divided into two teams. While students primarily worked within their assigned team, they also supported the other team in assistant roles on days when it was not their scheduled production day.

The LED-wall workflow provided a unified learning platform that integrated cinematography, lighting, real-time rendering, and staging. Students were able to prototype creative ideas and immediately observe their effects, gaining practical understanding of how virtual and physical elements combine in a live production. This environment reinforced the pipeline-centered learning emphasized throughout the course.

A key takeaway is the development of both ecosystem fluency and workflow literacy. Students engaged with camera calibration, stage lighting, asset preparation, and tools such as Unreal Engine, Blender, Nuke, Premiere, and Perforce, while learning how creative and technical decisions influence one another across the pipeline.

Working across multiple spaces and systems also required logistical coordination. Weekly plans, a shared Slack workspace, and collaboration with IT staff for equipment and network support made these operational dependencies visible, highlighting the importance of infrastructure and teamwork in large-scale media production.

ACKNOWLEDGMENTS

This work is supported by the Epic Games MegaGrant, RIT Previsualization and Virtual Production Curriculum Development with Unreal. Flip Phillips is supported, in part, by the National Science Foundation under Grant Award No. 2238180. We acknowledge the support of the MAGIC Center at RIT for hosting the course and providing the infrastructure that made this work possible. We are grateful to the MAGIC IT team for ensuring the reliability of the system infrastructure, and to the lab staff for facilitating smooth equipment check-in and check-out.

REFERENCES

Gregory Bennett, Hossein Najafi, and Lee Jackson. 2023. Pedagogical strategies for teaching Virtual Production pipelines. In *SIGGRAPH Asia 2023 Educator's Forum.* 1–10.

Alex Boutellier and Panos Raptis. 2023. Teaching Virtual Production: The Challenges of Developing a Formal Curriculum. *Film Education Journal* 6, 2 (2023), 69–81.

Manolya Kavakli and Cinzia Cremona. 2022. The virtual production studio concept–an emerging game changer in filmmaking. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR).* IEEE, 29–37.

Topher Maraffi. 2024. Metahuman Theatre: Teaching Photogrammetry and Mocap as a Performing Arts Process. In *ACM SIGGRAPH 2024 Educator's Forum.* 1–2.

Hossein Najafi, Jason Kennedy, Emily Ramsay, Masaya Todoroki, and Gregory Bennett. 2024. A Pedagogical Workflow for Interconnected Learning: Integrating Motion Capture in Animation, Visual Effects, and Game Design: Major/Minor curriculum structure that supports the integration of Motion Capture with Animation, Visual Effects and Game Design teaching pathways.. In *SIGGRAPH Asia 2024 Educator's Forum.* 1–7.

Gregg William Perkins and Santiago Echeverry. 2022. Virtual Production in Action: A Creative Implementation of Expanded Cinematography and Narratives. In *ACM SIGGRAPH 2022 Posters* (Vancouver, BC, Canada) *(SIGGRAPH '22).* Association for Computing Machinery, New York, NY, USA, Article 21, 2 pages. https://doi.org/10.1145/3532719.3543231

Jon Swords and Nina Willment. 2024. The emergence of virtual production–a research agenda. *Convergence* 30, 5 (2024), 1557–1574.